CYCLE 3 & ADVANCED | GRADES 9-12 | 3 HOURS

جامعة أبوظبي
Abu Dhabi University

Family AI Game Day

A family chooses the fun. Your app hosts it and judges fairly.

The challenge is a working website or app for family games inspired by culture, heritage or sports. Its AI observes real play through video and audio, applies the game rules and keeps score.

CYCLE 3

3 games: football + 2 original games

ADVANCED (CHAMPIONS)

5 games: football + 4 original games

The game collection includes both team and individual competition. Families can choose which game to play. The additional games are your team's own ideas.



1 CHOOSE A GAME

[Football]
[Your game 2]
[Your game 3]
[Choose]

2 PLAYERS & TEAMS

Players: P1, P2, P3, P4
Team A: P1 + P2
Team B: P3 + P4
[Save teams]

3 RULES & READY

Points: [game rules]
Time: [match length]
Tie: [tie-break rule]
[Everyone ready]

4 ROUND IN PLAY

Round 2 of 3 | P2's turn
Challenge: [shown here]
Time left: 00:30
A: 2 | B: 1

5 AI JUDGE RESULTS

[Upload / Record video]
Goal: Team A
[Game moment + reason]
[Apply AI results]

6 FINAL RESULTS

Team A wins: 3-2
[Match facts]
MVP: [player]
[Choose another game]

The human host (app user) sets up and runs the games

The app helps the host set up games, players, teams and challenges. It explains points, time limits, penalties, ties and winning. It manages turns, rounds, timers and scores, shows results with reasons, and lets the family choose another game.

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Your App Relies on an AI Judge

The AI watches and listens to the game, then uses the rules to judge it.



Football is a must + 2 (regular track) or 4 (advanced track) additional games

The AI follows the players and ball in the video. It spots goals and keeps score. It shows which team has the ball for the most time and which player has it for the most time. For this challenge, that player is the Most Valuable Player (MVP). These results come from what the AI can see in the game.

From game video to game updates

The app accepts an uploaded game video with audio, or records directly through a laptop or mobile camera and microphone. Either option is allowed. The camera angle clearly shows the action. Teams can wear different colours and numbered shirts. The AI finds facts such as who has the ball, whether a goal was scored or what answer was spoken. The app uses these facts to update scores, turns, rounds and results.

Two ways to connect the AI judge

The AI can work through the app, or separate Python code can process the video and audio and produce a results report. The report can be uploaded or copied into the app to update play. The app checks that the report came from your AI judge and was not changed, using a code or hash supplied by the AI judge.

Show the AI judge working

Your demo must show that the AI judging works. It shows real gameplay, what the AI finds in the video or audio, and how the app uses those results to update the score and move the game forward.

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Examples

These games are examples only and cannot be used in your entry. Your team must invent/choose other games. Be unique! Ask yourself: What games are suited for AI judging?

Heritage guessing game



Cup-stacking challenge



Ring-toss challenge



Copy-the-rhythm game



Your demo video

The participating students must demonstrate using and testing their own app and its AI. This is what we expect to see in the demo video.

The video may be longer than two minutes if the game needs more time to demonstrate. Please keep the video as short as possible.

The guides shared with you give more details about all deliverables and where and how to submit them.